Global Markets Daily: Can Japanese Savings Boost JGBs?

- Recent comments by Japanese finance minister Katayama and Prime Minister Takaichi suggest policymakers are considering ways to increase domestic purchases of Japanese assets, with two potential measures that could support JGBs: adjusting GPIF's asset allocation towards domestic assets, and allowing JGBs to be included in the tax-free NISA savings vehicle.
- Under GPIF's current strategy, there is roughly USD75bn of room to increase allocation to JGBs. While not particularly large relative to ongoing rebalancing flows, which are themselves generated by higher yields, a weaker currency, and domestic equity outperformance, a pro-active reallocation would carry greater signalling value than these mechanical flows.
- We estimate that shifting the domestic bond share from 27% to 31% would be worth around 3-6bp in 30y swap spreads, comparable to the 8bp round trip seen this week, though the pace of portfolio adjustment is likely to be slow given the domestic bond share has only fluctuated by 1-2ppts per year.
- Allowing tax-advantaged ownership of JGBs through NISA could increase demand, but low household allocations are more likely a function of historically low yields than tax status; as in Italy, rising retail ownership is typically associated with rising, not falling, yields.
- Both measures have the potential to increase demand for JGBs, but the majority of the impact is likely to come through signalling and positive feedback loops rather than outright purchases. We see more room for 20s and 30s to outperform alongside wider 20y and 30y swap spreads, but a sustainable decline in JGB volatility ultimately requires shifts to fiscal and monetary policy rather than supply, as most of the cheapening in the curve has been macro-led.

George Cole
+44(20)7552-1214 |
george.cole@gs.com
Goldman Sachs International

Isabella Rosenberg
+1(212)357-7628 |
isabella.rosenberg@gs.com
Goldman Sachs & Co. LLC

Can Japanese Savings Boost JGBs?

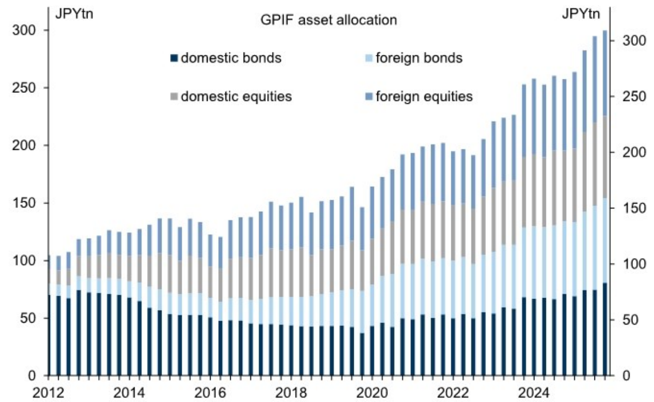
Recent comments by Japanese Finance Minister Katayama and Prime Minister Takaichi suggest policymakers in Japan are considering ways to increase domestic purchases of Japanese assets. JGBs rallied on the initial comments, with long-end JGBs outperforming in particular vs swaps. These moves have subsequently retraced, however, as the latest communication seemed to skew towards domestic equities rather than bonds.

In principle, the use of Japan's strong NIIP position could be used to bolster domestic assets and stabilise the currency, as domestic savings are redirected. As we will discuss below, this comes at some economic risk however, and without complementary macro shifts in policy and/or inflation, may not generate new equilibrium conditions for Japanese assets. Moreover, Japan already has among the lowest G10 foreign ownership shares of its government bonds, at around 14%, which leaves limited room for further domestic savings to be directed to JGBs. Nonetheless, two possibilities have been proposed to support domestic assets and JGBs: i) adjusting the asset allocation of the Government Pension Investment Fund (GPIF) towards domestic assets, and ii) allowing JGBs to be included in the Nippon Individual Savings Accounts (NISA), a tax-free savings vehicle offered to households. We investigate the potential implications of each.

GPIF

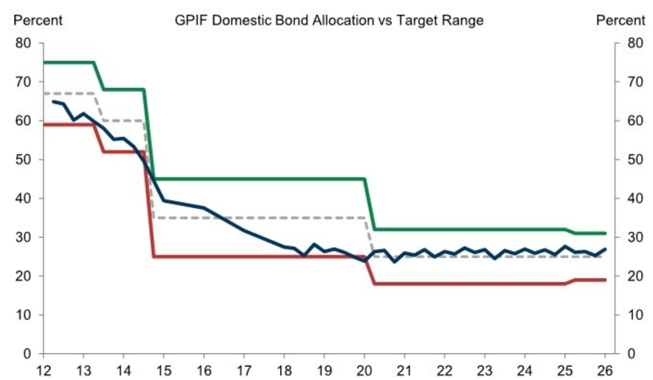
As of end-March 2026, GPIF manages around USD2.8tn in assets, allocated around a 25% benchmark for each of i) domestic bonds, ii) foreign bonds, iii) domestic equity and iv) foreign equity. Since 2014, GPIF holdings have gravitated away from domestic bonds to the roughly equal shares currently (Exhibit 1 & Exhibit 2). Current ownership represents around 5% of the JGB market, and just under 1% of the UST, Bund and Gilt market, among other G10s.

Exhibit 1: GPIF holdings have gravitated away from domestic bonds



Source: Goldman Sachs Global Investment Research, Haver Analytics

Exhibit 2: GPIF's current strategy targets a 25% allocation to domestic bonds, within a +/-6% range that has narrowed over time

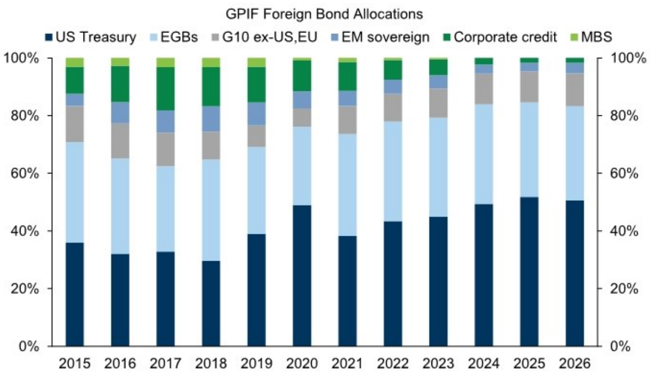


Source: Haver Analytics, Goldman Sachs Global Investment Research

From GPIF's annual data on individual bond holdings[1], we calculate JGBs typically represent around 80-90% of GPIF's domestic bond holdings, while USTs represent around 50% of foreign bonds (Exhibit 3). Since 2019 foreign bond holdings have moved further towards sovereign bonds and away from corporate credit, with USTs and EGBs over 80% of the total. The weighted average maturity of GPIF's sovereign holdings has been relatively steady, and close to or just above the overall stock in each market (Exhibit 4).

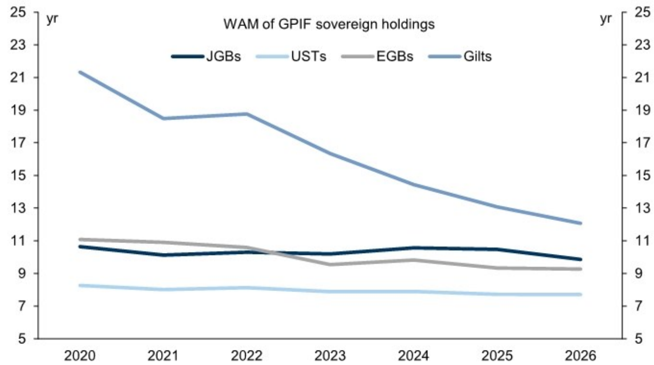
[1] GPIF publishes annual data on ISIN level holdings since 2019.

Exhibit 3: JGBs represent around 80-90% of domestic bond holdings, while USTs represent around 50% of foreign bond holdings



Source: GPIF, Goldman Sachs Global Investment Research

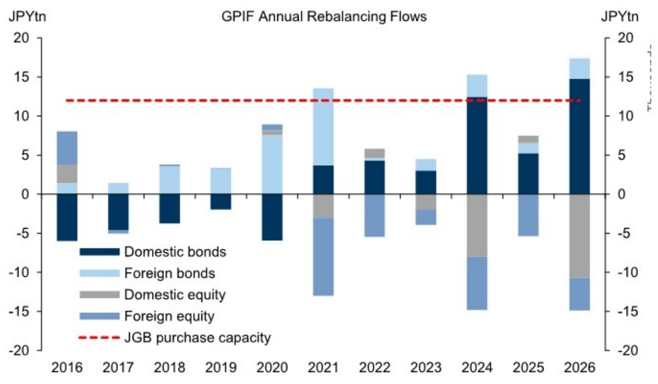
Exhibit 4: The WAM of GPIF's sovereign holdings has been relatively steady and close to the market, outside of Gilts



Source: Goldman Sachs Global Investment Research, Bloomberg

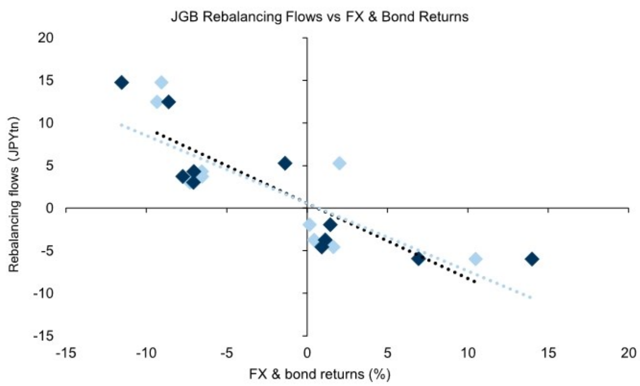
GPIF's current strategy (due for review in 2030) allows for flexibility around this 25% share—for domestic bonds, this is currently +/-6ppts, a share that has narrowed over time (Exhibit 2). Under the current strategy (also due for revision in 2030), there is roughly USD75bn of room to increase allocation to JGBs. But this is not particularly large in the context of GPIF's ongoing purchases in the market. Given strong investment returns from equity markets, and foreign assets in general, owing to a much weaker JPY in recent years, plus the underperformance of JGBs from higher yields, GPIF needs to re-balance its portfolio towards JGBs just to keep the share of domestic bonds stable over time (Exhibit 5). Recent rebalancing towards JGBs has actually been larger than the current capacity to expand JGB purchases. Of course, the ultimate market effect of a pro-active reallocation to JGBs would likely be more meaningful and have greater signalling value than rebalancing flows, which themselves are generated by higher yields and a weaker currency (Exhibit 6).

Exhibit 5: GPIF needs to re-balance its portfolio towards JGBs just to keep the share of domestic bonds stable over time



Source: Haver Analytics, GPIF, Goldman Sachs Global Investment Research

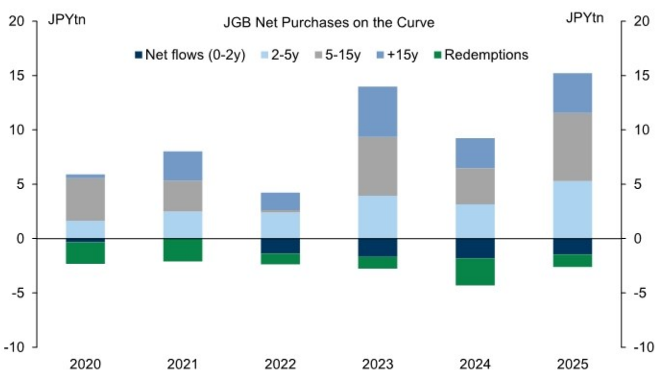
Exhibit 6: Rebalancing flows are generated by higher yields and a weaker currency



Source: Bloomberg, GPIF, Goldman Sachs Global Investment Research

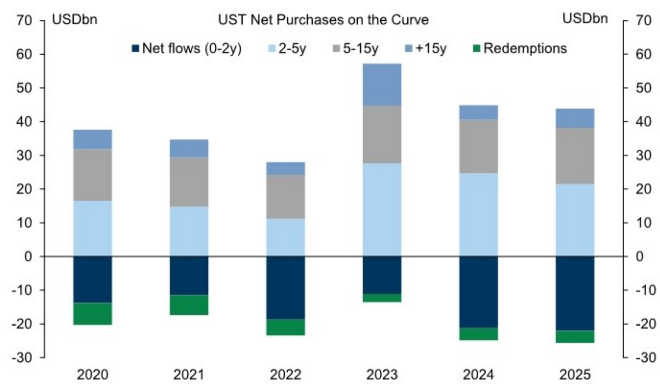
One possibility for increased flows to JGBs to have a more meaningful impact is a redirection towards the long-end. GPIF already does this to some extent—given the WAM of holdings is relatively constant (outside of Gilts), this requires a constant rebalancing within each curve towards longer maturities. By stripping out valuation changes due to price (yield) and currency movements in the case of foreign bonds, we see that on net, GPIF manages each curve by selling its shorter maturity bonds (sub-2y) and purchasing longer maturities—Exhibit 7 and Exhibit 8 show this for both JGBs and USTs. This shows at least some capacity to direct flows towards long-end bonds, which is where we have argued there is probably the most 'bang for the buck' for the JPY curve.

Exhibit 7: GPIF manages WAM by selling its shorter maturity bonds (sub-2y) and purchasing longer maturities in JGBs...



Source: GPIF, Goldman Sachs Global Investment Research

Exhibit 8: ...and its foreign bond holdings

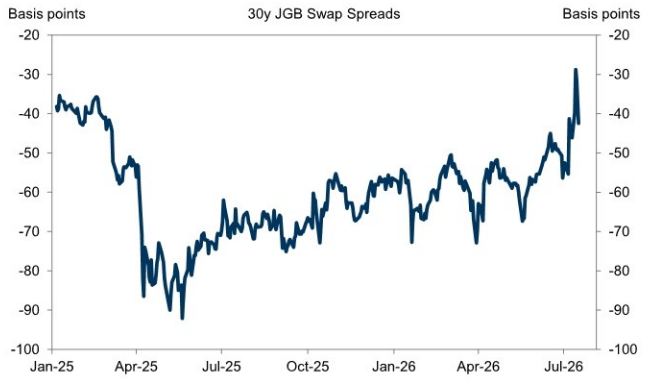


Source: GPIF, Goldman Sachs Global Investment Research

Although the sensitivities of JGB swap spreads to debt supply have historically been a little lower than in other markets, the recent volatility in JGBs points to the potential for increased local sensitivity to supply shifts. For USTs, Bunds and Gilts, our swap spread frameworks estimate that a 1ppt change in "free float" (the share of debt owned by the private sector) is worth around 1bp on 30y swap spreads, but the impact could be larger (e.g. 2-4bp) if flows are disproportionately directed into 30y bonds or catalyses more 'crowding in' from other domestic buyers beyond any direct GPIF flows. Shifting from the current domestic bond share of 27% to 31% (the maximum under the current strategy, and around USD75bn in addition JGB purchases) would represent around 3-6bp in 30y spreads—this compares with an 8bp round trip in 30y swap spreads this week (Exhibit 9).

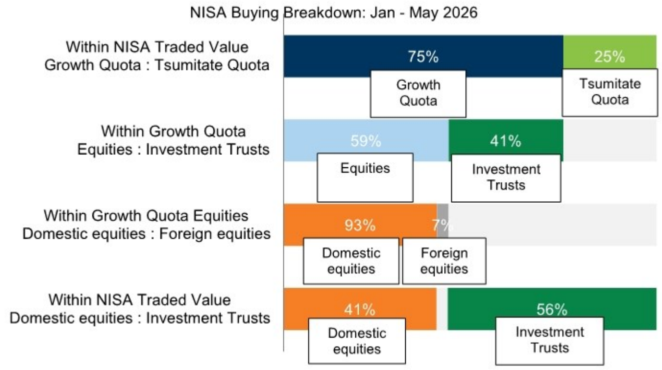
The volatile market reaction may reflect an extrapolation of potential GPIF reallocation to the broader pension industry. Of course, the size of the near-term reaction would also depend on the speed of adjustment but given the domestic bond share has only fluctuated by 1-2ppts per year, this shift is likely to be slow. And more meaningful shifts, such as the benchmark share allocated to domestic vs foreign assets, would likely need to wait several years. Less tangibly, the market may also be reacting positively to signs that the government is more attuned to risks in the bond market and searching for ways to address them. These measures are ultimately not a solution to the macro-led volatility in rates, but instead could function as a means to attenuate volatility from the current monetary and fiscal policy stance.

Exhibit 9: 30y swap spreads have moved by 8-10 basis points this week



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Exhibit 10: NISA's growth quota currently allows for investments in JGB funds, but not direct investment in JGBs



Source: JSDA, Goldman Sachs Global Investment Research

Given the relatively low ownership share of G10 bond markets (less than 1pp of the major bond markets), modest redirection of GPIF holdings away from foreign bonds to JGBs would have a small direct impact. Given the revealed behaviour to rebalancing by selling short maturity bonds, it is also not clear this would have a major curve impact. Of course, should it become a broader policy of domestic investors to divest or redirect away from foreign bonds, it could become more meaningful, but we do not consider this a key input for cross market or swap spreads.

NISA

The second measure that has been proposed is to make JGBs eligible in Nippon Individual Savings Accounts (NISA), a tax-free savings vehicle offered to households intended to support conversion of cash savings into long-term investments. Over 20mn accounts have been opened with over JPY71tn in traded value as of end-2025, indicating widespread usage. Within the "growth quota", which makes up 75% of NISA traded value, investors are able to allocate purchases to a broad range of assets, including individual domestic and foreign equities, ETFs, REITs, and active mutual funds (Exhibit 10). NISA's growth quota currently allows for investments in JGB funds, but not direct investment in JGBs. (The Tsumitate quota, in contrast, is restricted to low cost, diversified, equity mutual funds.) Direct ownership of JGBs across households is low, at about 4% of portfolio assets after stripping out NISA accounts.

Allowing for tax-advantaged ownership of JGBs could increase demand, but low allocations are likely a function of historically low yields than tax status. As we have shown in the case of Italy, increasing retail ownership is typically associated with rising, not falling, yields. The same is true of GPIF flows—as we described above, substantial rebalancing flows are a consequence of rising yields and JPY underperformance, rather than a driver of JGB outperformance. This suggests to us that while there may be modest swap spread effects from some redirection of domestic savings to JGBs, we think the route to lower yields and lower rate volatility is a macro one—some combination of tighter fiscal or monetary policy, and/or a downward shift in the inflation trajectory.

Both the GPIF and NISA measures have the potential to increase demand for JGBs, but their impact is likely to come through signaling and positive feedback loops rather than outright purchases. And we see the impact on other bond markets as even more limited. Given more supply-related cheapening at the very long end, we think 20s and 30s have more room to richen, alongside 20y and 30y swap spreads. The macro-led cheapening in 10s and the belly are less likely to be supported, as evidenced by persistent 10s30s flattening. Ultimately, the bulk of the cheapening in the curve has been policy-related, meaning that a sustainable decline in JGB volatility has to come from shifts to fiscal and monetary policy rather than supply.

TRADE IDEAS

Best Trade Ideas Across Assets
For pricing, charts, and a list of previous recommendations, please visit our Trade Ideas page.

1. Stay short SGD/MYR, opened January 24, 2026, at 3.13, with a target at 2.90 and a stop at 3.30, currently trading at 3.17.
2. Stay long TRY, NGN and KZT against the USD, as an equally weighted basket, opened February 18, 2026, at 0%, with a total return target of 7.5%, and a revised stop of 1.5%, currently trading at 4.2%.
3. Stay long 3y SOFR swap spread, opened 17 April 2026, at -22.6bp, with a revised target at -15bp inclusive of carry; and a revised stop at -20bp, currently trading at -18.3bp.
4. Stay short USD/EGP, opened 24 April 2026, at 0%, with a revised total return target at 12% and a revised stop at 5%, currently trading at 7.4%.
5. Stay long 5y Receivers on 3m 2s5s10s Receiver Fly (bp running), opened 09 May 2026, at 1bp, with a target at 10bp and a stop at -5bp, currently trading at 0bp.
6. 1y forward 2s10s GBP OIS steepeners, opened 29 May 2026, at 0.38, with a target at 0.55 and a stop at 0.25, currently trading at 0.31.
7. Stay short THB/INR, opened 12 June 2026, at 2.91, with a target at 2.70 and a stop at 3.05, currently trading at 2.86.
8. Stay long INR 30y bonds, opened 27 June 2026, at 7.34%, with a target at 6.90% and a stop at 7.65%, currently trading at 7.40%.
9. 2s10s NZD steepeners, opened 03 July 2026, at 72bp, with a target at 90bp and a stop at 62bp, currently trading at 75bp.
10. 2s10s CAD steepeners, opened 10 July 2026, at 53bp, with a target at 80bp and a revised stop at 46bp, currently trading at 54bp.
11. Stay short AUD/NZD, via 6m 1.1650 puts, opened 14 July 2026, at 1.2000, currently trading at 1.1950.
12. Go long NIFTY Banks vs. short NIFTY Pharma, opened 15 July 2026, at 100 in local currency (INR), with a target at 115 and a stop at 90, currently trading at 103.
13. Go short GBP/USD, opened 17 July 2026, at 1.3452. with a target of 1.3250 and a

stop of 1.3600.

Disclosure Appendix

Reg AC

We, George Cole and Isabella Rosenberg, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: George Cole Goldman Sachs International, Isabella Rosenberg Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints.

As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.